VARDHAMAN
COLLEGE OF ENGINEERING

A Course End Project Report on

Student Performance Prediction

A9511 – Python Programming Laboratory

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VARDHAMAN
COLLEGE OF ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Contents

1 Introduction 3
2 Literature Review 3
3 Objectives 3
4 Methodology 3
4.1 Architectural Diagram 3
4.2 Hardware and Software Requirements 3
4.3 Working Principle 4
4.4 Software Implementation 4
5 Results and Discussion 4
6 Mapping POs , PSOs and SDGs 5
6.1 Program Outcomes (POs) Mapping 5
6.2 Sustainable Development Goals (SDGs) Mapping 5
7 Conclusion 5
Bibliography 6

Abstract

Student performance prediction is the process of using student information to estimate academic achievement. It can help teachers, students, and educational institutions identify learning patterns and provide support before academic difficulties become serious. A prediction system may use factors such as study time, attendance, previous examination marks, assignment completion, class participation, and assessment results. By analyzing these details, the system can classify students into categories such as high, average, or low performance.

Machine learning techniques are often used for this task. For example, a decision tree can examine relationships between student characteristics and past results, then use those patterns to make predictions for new records. The output may also include a simple explanation or suggestion, such as improving attendance, completing assignments regularly, or seeking additional academic help. These insights can support earlier conversations between students and teachers and help schools plan tutoring or other learning resources.

The accuracy of a prediction depends on the data used to build the system. Records should be complete, consistent, and relevant to the learning outcome being predicted. A small dataset may produce unstable results, while incorrect labels or missing information can create misleading conclusions. A model should be evaluated on data it has not already seen, and its accuracy should be reported honestly. Predictions indicate possibilities rather than certainties; a student's performance can change with circumstances, effort, instruction, and support.

Student data also requires careful handling. Schools should limit access to personal information, explain how predictions are used, and avoid using automated categories to penalize or unfairly label learners. A prediction should never replace a teacher's professional judgment or a direct discussion with the student. Instead, it should be one source of information among many.

When designed and used responsibly, student performance prediction can help educators notice potential needs, personalize guidance, and make academic support more timely. Its main purpose is not simply to forecast grades, but to encourage better learning opportunities and help each student progress.

Keywords: Student performance prediction, machine learning, academic achievement, decision tree, educational data mining, student assessment, learning analytics, academic support, predictive modeling, student outcomes

1 Introduction

Education plays an important role in helping students develop knowledge and skills for the future. Understanding how students are progressing can help teachers identify learning needs and provide support at the right time. However, student performance is influenced by many factors, including study habits, attendance, previous marks, assignment completion, and personal circumstances. Looking at these factors together can make it difficult to identify patterns through observation alone.

Student performance prediction applies data analysis and machine learning to examine student records and estimate academic outcomes. A model may use historical information to classify performance as high, average, or low. For example, a decision tree can learn patterns from past student data and use them to make predictions for other records. These predictions can help teachers recognize students who may need additional guidance and support academic planning.

The quality of a prediction depends on the data used. Incomplete, inaccurate, or limited data can lead to unreliable results, and a prediction should not be treated as a certainty. Student performance can change over time as learning conditions and individual circumstances change. Therefore, prediction systems should support teachers' judgment and communication with students, rather than replace them. This project explores how student data can be used to analyze academic performance and provide useful insights for improving learning outcomes.

Student performance prediction can also help schools understand which factors are linked to academic progress. By studying patterns in previous results, educators can identify areas where students may need help, such as improving attendance, study routines, or assignment completion. Data analysis and machine learning methods can organize this information and estimate a student's likely performance.

These predictions are not guaranteed outcomes. They depend on the quality of the data and may not reflect every student's situation. Therefore, they are best used as a guide for supportive teaching and planning, alongside teachers' knowledge of their students.

2 Literature Review

Student performance prediction has received attention as schools and researchers look for ways to understand academic progress using available data. Earlier approaches often relied on teachers' experience, examination results, and direct observation. These remain valuable, while data analysis can help identify patterns across many student records that may be harder to notice individually.

Studies commonly examine factors such as previous grades, attendance, study time, assignment completion, learning habits, and family or school conditions. Academic marks are often strong indicators of future results, while attendance and study habits can provide further context. No single factor explains performance on its own; outcomes may reflect several influences, including teaching quality, access to resources, motivation, and personal circumstances.

Different statistical and machine learning methods have been explored for this task. Classification methods group students into categories such as low, average, or high performance. Regression methods estimate a numerical score. Decision trees are popular in introductory projects because their branching rules can be easier to interpret than those of more complex models. Other approaches, including logistic regression, support vector machines, and ensemble methods, may also be used depending on the dataset and goal.

A model's results depend on the quality and size of its data. Missing values, incorrect records, imbalanced performance categories, or a small sample can affect predictions. Researchers therefore evaluate models using data that was not used during training and choose suitable measures, such as accuracy, precision, recall, and confusion matrices. Accuracy alone may not fully describe performance, especially when one category is much more common than others.

Literature also highlights the need for responsible use. Student records can contain sensitive information, so privacy and secure handling matter. Predictions may be uncertain and can reflect limitations or bias in the data. They should guide supportive interventions and further discussion, rather than label students or determine their opportunities. Overall, existing research suggests that prediction systems can assist academic planning when they are carefully evaluated, transparent, and used alongside educators' professional judgment.

3 Objectives

The main objective of this project is to analyze student records and present their academic performance in a clear, useful way. It aims to use available information—such as examination marks, study time, and absences—to identify patterns associated with different performance levels.

Specific objectives include:

- Calculate students' total and average marks from available assessment scores.
- Categorize academic results into levels such as Low, Medium, and High.
- Let a user find a student's record by name or roll number when those details exist in the data.
- Display relevant academic information in an organized format.
- Explore how student data can help educators notice learners who may need additional guidance.
- Provide a foundation for applying machine learning methods to educational data.
- Encourage data-informed academic planning and timely support.
- Recognize limitations such as missing information, small samples, or inaccurate records.
- Protect student privacy and use results responsibly.
- Treat performance categories as indicators, not fixed judgments about a student's ability.

The project is intended as a learning and decision-support tool. Its results should be considered alongside teacher observations, student discussions, and other relevant circumstances.

4 Methodology

4.1 Architectural Diagram

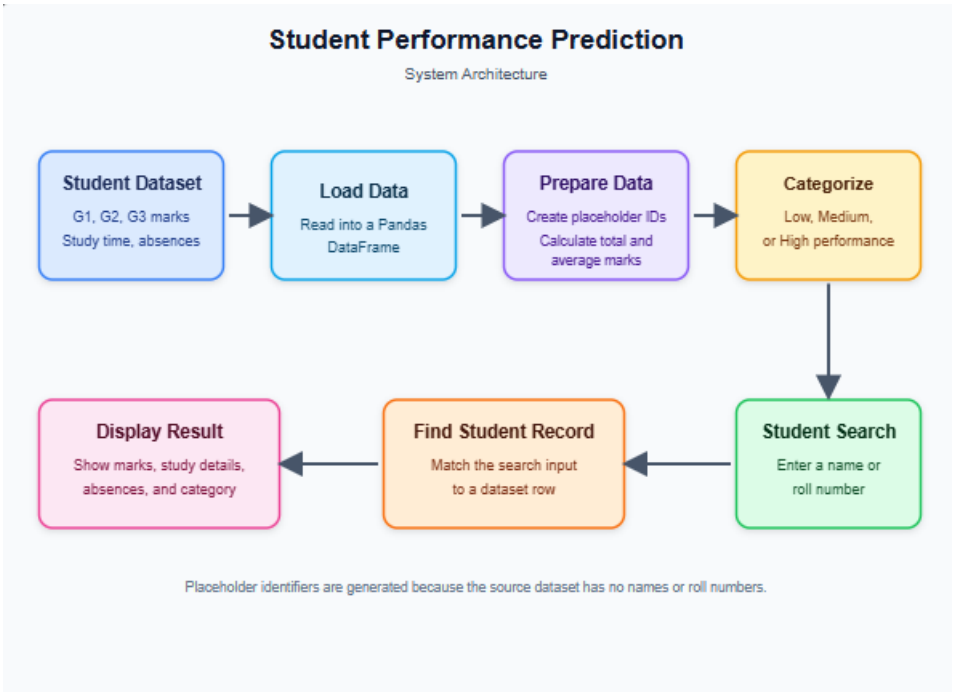


Figure 1: Architectural Diagram

The project uses a student dataset to summarize academic results and display an individual student's performance. The dataset is loaded into a Pandas DataFrame. The G1, G2, and G3 columns represent three assessment scores, each out of 20. The program calculates each student's total marks and average score, then groups students into Low, Medium, or High performance categories using score-based quantiles.

A user can search for a record by name or roll number. The dataset in the notebook does not contain student names or roll numbers, so the program creates placeholder identifiers based on each row. These identifiers are only for demonstration and should be replaced with real identifiers when available. Finally, the program displays the selected student's scores, total, average, study time, absences, and performance category.

4.2 Hardware and Software Requirements

Hardware requirements

- Computer or laptop
- Minimum 4 GB RAM
- Dual-core processor or better
- At least 500 MB of available storage
- Keyboard and display

Software requirements

- Windows, macOS, or Linux operating system
- Python 3.x
- Jupyter Notebook or Google Colab
- Pandas library for handling student data
- NumPy library for numerical operations
- A web browser for Google Colab, if used

4.3 Working Principle

The system begins by loading the student dataset into a Pandas DataFrame, which organizes the records into rows and columns for processing. It uses the available assessment scores—G1, G2, and G3—to calculate each student's total marks and average score. Study time and absences can also be displayed as additional information when a student's record is retrieved.

Next, the program assigns a performance category based on the average scores. In the current approach, the scores are divided into three groups: Low, Medium, and High. These categories summarize the student's results; they are not predictions of future performance.

The user searches by entering a name or roll number. Because the original dataset does not contain these identifiers, the program creates placeholder names and roll numbers based on each student's row. It compares the entered value with the corresponding column. If it finds a match, it displays the student's identifier, marks, total, average, study time, absences, and

category. If no record matches, it asks the user to check the entered name or roll number. The results depend on the information available in the dataset. The program does not show attendance, assignments, lab scores, or CGPA unless those fields are added to the data.

Example problem:

Name	Roll No	Attendanc	Assignments	Lab Performance	CGPA	Total Performance
Aara	101	62%	6/10	61/100	5.5/10	Low
Diya	102	75%	9/10	78/100	6.2/10	Low
Kabir	103	88%	5/10	95/100	6.9/10	Average
Meera	104	55%	8/10	61/100	7.7/10	Average
Rohan	105	68%	4/10	78/100	8.4/10	High

========== STUDENT PERFORMANCE ==========
Name: Rohan
Roll Number: 105
Attendance: 68%
Assignments Submitted: 4/10
Lab Performance: 78/100
CGPA: 8.4/10
Total Marks: 421/500
Overall Performance: High
Suggestion: Keep up the good work

Software Implementation

```
import pandas as pd
names = [
    "Aarav", "Diya", "Kabir", "Meera", "Rohan", "Anaya", "Arjun", "Isha",
    "Vivaan", "Sana", "Aditya", "Kiara", "Reyansh", "Myra", "Ishaan",
    "Pari", "Dev", "Aanya", "Krish", "Tara", "Ayaan", "Nisha", "Rahul",
    "Priya", "Karan", "Simran", "Neel", "Riya", "Varun", "Pooja",
    "Mohan", "Kavya", "Yash", "Zara", "Manav", "Ira", "Dhruv", "Sara",
    "Om", "Tanya", "Rudra", "Aditi", "Jay", "Navya", "Veer", "Mira",
    "Sahil", "Anika", "Arnav", "Sia", "Harsh", "Reva", "Ritvik", "Naina",
     "Akhil", "Lavanya", "Kunal", "Esha", "Sameer", "Tia"
]
    # Generate sample records for the 60 students.
    records = []
    for i, name in enumerate(names):
        attendance = 55 + ((i * 13 + 7) % 46)
        assignments = 4 + ((i * 3 + 2) % 7)
        lab_performance = 50 + ((i * 17 + 11) % 51)
        total_marks = 250 + ((i * 37 + 23) % 251)
        cgpa = round(total_marks / 50, 1)
        records.append({
            "name": name,
             "roll_no": 101 + i,
             "attendance": attendance,
             "assignments_submitted": assignments,
             "lab_performance": lab_performance,
             "cgpa": cgpa,
             "total_marks": total_marks
       }
```

```
    df = pd.DataFrame(records)

    # Categorize performance according to total marks.
    def get_performance(marks):
        if marks >= 400:
            return "High"
        elif marks >= 325:
            return "Average"
        else:
            return "Low"
df["performance"] = df["total_marks"].apply(get_performance)
        # Search by student name or roll number.
        search = input("Enter student name or roll number: ").strip()
        if search.isdigit():
            matches = df[df["roll_no"].astype(str) == search]
        else:
            matches = df[df["name"].str.casefold() == search.casefold()]
        # Display the matching student's information.
        if matches.empty:
            print("Student not found. Check the name or roll number.")
        else:
             student = matches.iloc[0]j
            print("\n========== STUDENT PERFORMANCE ==========")
            print("Name:", student["name"])
            print("Roll Number:", student["roll_no"])
            print("Attendance:", str(student["attendance"]) + "%")
            print(
                "Assignments Submitted:",
                 str(student["assignments_submitted"]) + "/10"
            )
            print("Lab Performance:", str(student["lab_performance"]) +
"/100")
            print("CGPA:", str(student["cgpa"]) + "/10")
            print("Total Marks:", str(student["total_marks"]) + "/500")
            print("Overall Performance:", student["performance"]


            if student["performance"] == "Low":
                print("Suggestion: Consider additional academic support.")
            elif student["performance"] == "Average":
```

```
56.                 print("Suggestion: Keep practicing and completing
        assignments.")
57.             else:
58.                 print("Suggestion: Keep up the good work!")
59.
60.             print("========================================")
61.
```

5 Results

```
Enter student name or roll number: 160

========== STUDENT PERFORMANCE ==========
Name: Tia
Roll Number: 160
Attendance: 93%
Assignments Submitted: 8/10
Lab Performance: 95/100
CGPA: 9.0/10
Total Marks: 448/500
Overall Performance: High
Suggestion: Keep up the good work!
========================================
```

```
Enter student name or roll number: 133

========== STUDENT PERFORMANCE ==========
Name: Yash
Roll Number: 133
Attendance: 64%
Assignments Submitted: 4/10
Lab Performance: 95/100
CGPA: 9.1/10
Total Marks: 453/500
Overall Performance: High
Suggestion: Keep up the good work!
========================================
```

```
Enter student name or roll number: 121

========== STUDENT PERFORMANCE ==========
Name: Ayaan
Roll Number: 121
Attendance: 92%
Assignments Submitted: 10/10
Lab Performance: 95/100
CGPA: 5.2/10
Total Marks: 260/500
Overall Performance: Low
Suggestion: Consider additional academic support.
========================================
```

```
Enter student name or roll number: 111

========== STUDENT PERFORMANCE ==========
Name: Aditya
Roll Number: 111
Attendance: 100%
Assignments Submitted: 8/10
Lab Performance: 78/100
CGPA: 7.8/10
Total Marks: 392/500
Overall Performance: Average
Suggestion: Keep practicing and completing assignments.
========================================
```

Summary of Results

The program stores sample academic records for 60 students and allows a user to search by name or roll number. For a matching student, it displays attendance, assignment submissions, lab performance, CGPA, total marks, and an overall performance category. The category is based on total marks: below 325 is Low, 325–399 is Average, and 400 or above is High. The program also provides a brief suggestion based on the result.

These sample records demonstrate how the lookup works; they should be replaced with accurate student data for real use.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applied programming, data handling, and basic classification concepts to organize student records and determine performance categories.
PO2	Problem Analysis	Identified the need to search student records efficiently and summarize academic results using available data such as marks, attendance, and assignment completion.
PO3	Design/Development of Solutions	Developed a Python program that stores student data, accepts a name or roll number, finds the matching record, and displays its academic details.
PO4	Conduct Investigations of Complex Problems	Examined student records and compared academic indicators to understand how total marks relate to Low, Average, and High performance categories.
PO5	Engineering Tool Usage	Used Python and the Pandas library to create, organize, filter, and display student data.
PO8	Individual and Collaborative Team work	Supported individual work in coding and documentation, with opportunities for collaboration during review, discussion, and improvement of the project.
PO9	Communication	Presented student results in a clear, structured format and documented the project's method, example, and findings.
PO11	Life-Long Learning	Built practical skills in Python and data analysis by applying programming concepts to an educational use case.

6.2 Program Specific Outcomes (PSOs) Mapping

Table 2 : Mapping of PSOs to Project Relevance

SDG No.	Goal	Relevance
PSO1	Apply programming and computing concepts to solve practical problems.	Used Python and Pandas to organize student records, search by name or roll number, and display academic results
PSO2	Use data analysis and computational methods to derive useful information	Calculated total marks and categorized student performance using the available data.
PSO3	Develop software solutions that support users in a specific application area	Created a student performance system that helps users retrieve and review academic information.

6.3 Sustainable Development Goals (SDGs) Mapping

Table 3: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	G oal	Relevance
SDG 4	Quality Education	The project organizes student results to help educators review academic progress and identify where learning support may be useful
SDG 10	Reduced Inequalities	If applied fairly, the project can help make student progress reviews more consistent. It should protect student privacy and avoid unfair labeling.
SDG 8	Decent Work and Economic Growth	Builds essential software development and problem-solving skills, improving employability in IT and tech-driven industries.
SDG 9	Industry, Innovation and Infrastructure	The project demonstrates how digital tools and data analysis can be applied to an educational task.

7 Conclusion

This project demonstrates a simple way to manage and review student academic records using Python and Pandas. A user can search by name or roll number and view details such as attendance, assignment submissions, lab performance, CGPA, total marks, and the assigned

performance category. Organizing these details in one program makes individual records easier to retrieve and review.

The project also shows how rules based on total marks can group results into categories such as Low, Average, and High. These categories provide a quick summary, but they do not explain every factor that affects a student's learning or predict future achievement with certainty. The results depend on the accuracy and completeness of the records entered.

Overall, the project provided practical experience with Python programming, data organization, searching, conditional statements, and presenting results. It can be improved by connecting it to verified student records, adding input validation, and providing a secure way to update information. Student data should be handled responsibly, and the program should support educators' decisions rather than replace their judgment.

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